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Stressing schizotypy: The modulating role of stress-relieving behaviours and intellectual capacity on functional hemispheric asymmetry

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Relative cognitive impairments are common along the schizophrenia spectrum reflecting potential psychopathological markers. Yet stress, a vulnerability marker in schizophrenia (including its spectrum), is likewise related to cognitive impairments. We investigated whether one such cognitive marker (attenuated functional hemispheric asymmetry) during stressful life periods might be linked to individuals' schizotypal features or rather to individuals' stress-related experiences and behaviours. A total of 58 students performed a left hemisphere dominant (lateralised lexical decisions) and right hemisphere dominant (sex decisions on composite faces) task. In order to account for individual differences in stress sensitivity we separated participants into groups of high or low cognitive reserve according to their average current marks. In addition, participants filled in questionnaires on schizotypy (short O-LIFE), perceived stress, stress response, and a newly adapted questionnaire that enquired about potential stress compensation behaviour (elevated substance use). The most important finding was that enhanced substance use and cognitive disorganisation contributed to a right and left hemisphere shift in language dominance, respectively. We discuss that (i) former reports on right hemisphere shifts in language dominance with positive schizotypy might be explained by an associated higher substance use and (ii) cognitive

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disorganisation relates to unstable cognitive functioning that depend on individuals' life circumstances, contributing to published reports on inconsistent laterality-schizotypy relationships.

Keywords: Stress; Drug dependence; Schizophrenia spectrum; Cognitive reserve; Hemispheric asymmetry.

Relative cognitive impairments are common in patients with psychosis as well as in less-severely affected individuals along the schizophrenia spectrum. This includes individuals with a schizotypal personality disorder (for an overview see Reichenberg & Harvey, 2007) and those scoring high on schizotypy questionnaires (Burch, Hemsley, & Joseph, 2004; Krabbendam, Myin-Germeys, Hanssen, & van Os, 2005; Laurent et al., 2001; Poreh, Ross, & Whitman, 1995). For instance, higher cognitive functions that rely on the frontal lobes (e.g., working memory, fluency) seem impaired along the schizophrenia spectrum, including schizotypy (Laurent et al., 2001; Reichenberg & Harvey, 2007; Tsakanikos & Claridge, 2005). Supporting the likely brain correlates of such behavioural findings, the frontal lobes have long been discussed as a target area of psychopathology along the schizophrenia spectrum (Buchsbaum et al., 2002).

Most relevant to the present study are those reports that link common patterns of functional hemispheric asymmetry to intact frontal and temporal lobe functioning (Rossell, Bullmore, Williams, & David, 2001). In particular, the common hemispheric asymmetry pattern seems to be disrupted along the schizophrenia spectrum with reduced left hemisphere dominance for language or a reduced right hemisphere dominance for face processing reported from patients with schizophrenia (Bleich-Cohen, Hendler, Kotler, & Strous, 2009; Kucharska-Pietura, David, Dropko, & Klimkowski, 2002; Mitchell & Crow, 2005; Phillips & David, 1997; Sommer, Ramsey, & Kahn, 2001) and healthy individuals with elevated self-reported schizotypy (Broks, 1984; Brugger, Gamma, Muri, Schafer, & Taylor, 1993; Løberg et al., 2006; Mason & Claridge, 1999; Mohr et al., 2005; Suzuki & Usher, 2009). It is considered that cognitive impairments (such as those specified here) might be a reflection of the illness, or even be a marker of the schizophrenia spectrum (Crow, 2000; Noguchi, Hori, & Kunugi, 2008).

In addition to the above, the stress response system seems to be modulated along the schizophrenia spectrum as well. First, the hypothalamic-pituitary-adrenal (HPA) axis, one of the primary neural systems triggered by stress exposure, may be hyperactive in psychotic patients (Walker, Mittal, & Tessner, 2008). Second, increases in the stress-hormone cortisol can elevate psychotic symptoms in schizophrenia patients (for overviews see Walker & Diforio, 1997; Walker et al., 2008). Drugs that

increase cortisol serum levels induced mild psychotic symptoms and perceptual disturbances in healthy individuals (D'Souza et al., 2006). Third, an elevated stress response has also been found in schizotypy (Soliman et al., 2008; see also van Winkel, Stefanis, & Myin-Germeys, 2008), which may explain why individuals scoring high on positive schizotypy show sensitivity to threatening (stressful) stimuli (Fisher et al., 2004).

Stress has been linked not only to the schizophrenia spectrum, but also to the cognitive functions associated with it. For instance, cortisol administration disrupts verbal memory functions (Newcomer et al., 1999). Naturalistic stressors such as exam stress cause increased cortisol levels in some studies (Lucini, Norbiato, Clerici, & Pagani, 2002; Morgan, Rasmussen, Pietrzak, Coric, & Southwick, 2009), although not in others (Vedhara, Hyde, Gilchrist, Tytherleigh, & Plummer, 2000). Naturalistic stressors associate with shifts in hemispheric functioning such that relatively greater left frontal EEG activity during low examination stress, shifts to relatively greater right frontal activity during high examination stress (Lewis, Weekes, & Wang, 2007; see also Gruzelier & Phelan, 1991). This shift is associated with increasing health complaints (Lewis, Weekes, & Wang, 2007).

Given these cross-links between the schizophrenia spectrum, stress, and cognition, we investigated whether shifts in hemispheric asymmetry would be modulated by elevated stress rather than by participants' self-reported psychotic and psychotic-like thinking (Broks, 1984; Brugger et al., 1993; Løberg et al., 2006; Mason & Claridge, 1999; Mohr et al., 2005; Suzuki & Usher, 2009). We also accounted for individual differences in stress responses and evaluations by asking about stress-related behaviours/experiences and cognitive reserve for the following reasons. Stress-related behaviours and experiences such as smoking (Finkelstein, Kubzansky, & Goodman, 2006) and drug use (Goeders, 2003) have been found to increase in times of elevated stress including periods before/during exams (Kassel, Stroud, & Paronis, 2003; Ng & Jeffery, 2003; Steptoe, Wardle, Pollard, Canaan, & Davies, 1996; Umberson, Liu, & Reczek, 2008) because they might have stress-alleviating properties (henceforth referred to as compensatory behaviours). Such compensatory behaviours may influence cognition (e.g., Scarmeas & Stern, 2003), and seem to occur more frequently along the schizophrenia spectrum (Cantwell et al., 1999; Esterberg, Jones, Compton, & Walker, 2007; Kavanagh, McGrath, Saunders, Dore, & Clark, 2002; Regier et al., 1990). Cognitive reserve (also referred to as cognitive resilience, and sometimes linked to IQ) can be understood as an efficient dynamism with which the brain balances out threatening situations and challenges in order to maintain efficient cognitive functioning (Katzman et al., 1989; Stern, 2002). Cognitive reserve may protect from adverse situations in psychiatric illnesses (MacCabe & Murray, 2004; Moore et al., 2007; Sørensen et al., 2010) including post-traumatic stress disorder (for an overview see Buckley, Blanchard, & Neill, 2000).

In the present study we investigated whether uncommon patterns of hemispheric asymmetry might relate to an abnormal or overactive stress response rather than to individuals' self-reported schizotypy. We assessed whether reduced hemispheric asymmetry in a left hemisphere dominant (lateralised lexical decisions) and right hemisphere dominant (sex decisions in visual composite faces) task is modulated by individuals' experienced stress/compensatory behaviours and cognitive reserve rather than by their self-reported schizotypy. If stress is a more important predictor of hemispheric asymmetry than schizotypy, we expect that higher stress levels and higher stress-related compensatory behaviours will relate to a reduced hemispheric asymmetry for function, in particular in those with less cognitive reserve (relatively lower marks).

METHOD

Participants

This native English-speaking student sample ($n = 58$; males $n = 16$) reported normal or corrected to normal vision, and was right-handed according to a standardised handedness questionnaire (Oldfield, 1971; for scoring criteria see also Kita, de Condappa, & Mohr, 2007). The sample consisted of MSc students ($n = 29$) and BSc students at the University of Bristol. Their mean age (years, $\pm SD$) was 20.33 (± 2.35). Participants took part to obtain course credit or were recruited via a local Internet advertising system. The study was approved by the local ethics committee. All participants provided written informed consent prior to participation. As indicated by self-report, none of the participants reported a previous history of psychiatric or neurological illness (Mohr, Landis, & Brugger, 2006).

Self-report questionnaires

Perceived stress scale. This 14-item scale (S. Cohen, Kamarck, & Mermelstein, 1983) measures global levels of perceived stress tapping into the key components of perceived stress, namely how uncontrollable, unpredictable, and overloading (Averill, 1973; Houston, 1972; Monat, Averill, & Lazarus, 1972; Schulz, Kirschbaum, Prussner, & Hellhammer, 1998) respondents found their lives in the past month. Participants answer on a 5-point Likert scale ranging from "never", "almost never", "sometimes", "fairly often", and "very often". After coding all items so that higher scores indicate higher perceived stress levels, scores range from 0 to 56. The perceived stress scale has adequate internal validity, construct validity, and test-retest reliability (S. Cohen et al., 1983; Ramirez & Hernandez, 2007). Normative values can be found in Ramirez and Hernandez (2007).

Stress response inventory. This 39-item stress response inventory (Koh, Park, Kim, & Cho, 2001) measures emotional, somatic, cognitive, and behavioural stress responses tapping into different stress-response domains such as tension (6 statements), aggression (4 statements), somatisation (3 statements), anger (6 statements), depression (8 statements), fatigue (5 statements), and frustration (7 statements). People indicate how much each statement applied to them in the past month on a 5-point Likert scale with the response options being “Not at all” (scored “0”), “Somewhat” (scored “1”), “Moderately” (scored “2”), “Very much” (scored “3”), or “Absolutely” (scored “4”). After coding all items so that higher scores indicate a higher stress response, scores range from 0 to 156. The stress response inventory is considered to have adequate internal consistency and test–retest reliability, and normative values for different populations can be found in Koh et al. (2001).

Short O-LIFE. The short O-LIFE questionnaire (Mason, Linney, & Claridge, 2005) is a validated 43-item self-report questionnaire assessing schizotypy in terms of four dimensions. Positive schizotypy is assessed by 12 items pertaining to Unusual Experiences (UnEx, maximum score 12), negative schizotypy is assessed by 10 items pertaining to Introverted Anhedonia (IntAn, maximum score 10), and Cognitive Disorganisation is assessed by 11 items (CogDis, maximum score 11). Finally, 10 items assess Impulsive Nonconformity (Imp, maximum score 10), which does not represent a schizotypy dimension (Mason, Claridge, & Jackson, 1995), but will be accounted for in the present study because of the strong link between impulsivity and addiction (Crews & Boettiger, 2009). For each item participants have to indicate whether the statement is true or false. The number of positive responses (some items are reverse formulated) is summed so that higher scores indicate higher schizotypy. Normative values can be found in Mason et al. (2005) and the scale has shown good internal consistency as well as high correlations with the original O-LIFE questionnaire (Mason, et al., 1995, 2005).

Compensatory behaviours. This adapted questionnaire (see Appendix) is based on Ogden, Veale, and Summers' (1997) “Exercise Dependence Questionnaire”, and is aimed at assessing the severity or extent of substance-based compensatory behaviours (henceforth referred to as substance-CB, 16 items). To avoid stigmatisation and reduce associations with clinically relevant dependence, ethical concerns, and socially desirable answers, we did not ask for drug use or drug addiction, but focused on the need or habit to consume a given substance (see the questionnaire instructions in the Appendix). Nevertheless, the items assess a variety of life-domains commonly affected by clinically relevant dependency, such as “interference with social/family life”

(3 statements), “withdrawal symptoms” (3 statements), “insight into problem” (2 statements), “positive reward” (3 statements), “loss of control” (2 statements), and behaviour salience (3 statements). On a 5-point Likert scale, individuals indicate how often they think each statement applied to them in the past month, with response options being “Never” (scored “0”), “Sometimes” (scored “1”), “Often” (scored “2”), “Nearly always” (scored “3”), and “Always” (scored “4”). Thus scores range from 0 to 64 with a higher score indicating a higher “dependence” on substance-CB.

Cognitive reserve. To estimate cognitive reserve, participants indicated their current average mark. Answers were provided in percentages (0–100% scale), where 50% is a pass, 70–100% (for taught components and dissertations, respectively) is a first or distinction. Higher scores indicate higher marks and by inference higher cognitive reserve. The link between higher marks and higher IQ/cognitive reserve is well established (see Buckley et al., 2000, for overview; see also Spinath, Harald Freudenthaler, & Neubauer, 2010; Stern, 2002).

Hemi-field studies¹

We assessed hemispheric asymmetry with two lateralised lexical and facial decision tasks used previously (Herzig, Tracy, Munafò, & Mohr, 2010). Details can be found in that reference, and so we will only briefly describe the tasks here.

Lateralised Lexical Decision Task (LDT). The stimulus material consisted of 24 abstract words and 72 pronounceable nonwords. The stimuli consisted of four- and five-letter words and were matched for neighbourhood and CELEX frequency (Coltheart, 1981). Each word was matched with a nonword of the same length. The remaining nonwords were matched to result in an additional set of nonword pairs. There were 72 trials with three 24-trial conditions (word left/nonword right, nonword left/word right, and nonword/nonword as the control condition). Participants were instructed to indicate whether they saw a meaningful English word on the left or right, or did not see a meaningful English word at all, by pressing the shift key ipsilateral to the word with the index finger, or space bar with both thumbs if they did not see a meaningful string of letters on the screen. Prior to the experimental task each participant undertook a practice block consisting of 10 trials with words and nonwords not used in the experimental trial. The order of the stimuli was randomised within blocks and between participants, and trials were doubled according to suggestions by Hunter and Brysbaert (2008), resulting in a total of 144 trials. We assessed the number of correct

lexical decisions and the mean reaction times for correct lexical decisions for the left (LVF) and right (RVF) visual field separately.

Lateralised Facial Decision Task (FDT). Participants were presented with emotionally neutral facial stimuli against a grey background on the computer screen. In the control condition 20 whole faces (= WF, 10 male, 10 female) were presented. From these, 20 sexually dimorphic composite faces were constructed with an equal number of female and male half-faces appearing in each visual field. These composite faces were also presented mirror-reversed, resulting in 40 composite faces. Participants were instructed to press one of the two shift keys to indicate the sex of the picture. The allocation of face sex to the two shift keys was counterbalanced between participants. Prior to the test trials participants were presented with a practice block of 10 trials consisting of two whole faces and eight composite faces that were not included in the experimental trials. The order of stimuli was randomised within blocks and between participants, and trials were doubled according to suggestions by Hunter and Brysbaert (2008), resulting in 120 trials in total. We assessed the number and response time of facial decisions towards the left visual field (LF decisions) and right visual field (RF decisions).

Data cleaning and analysis

We excluded individual response latencies that were faster than $2 \times SD$ from the individual means of the different conditions in the LDT and FDT. To test for differences in performance as a function of cognitive reserve, we split the whole group into a group of high and low cognitive reserve at the median current average mark. Using these measures we calculated separate 2×2 mixed-samples ANOVAs with visual field (LVF, RVF) as the repeated factor and cognitive reserve (high, low) as the independent factor on (i) mean reaction times of correct responses (LVF/RVF in the LDT), and (ii) percent correct responses (LVF/RVF in the LDT). For the FDT (see Herzig et al., 2010, for detailed rationale) we performed a mixed-samples ANOVA on percent correct (WF) and percent LF decisions (composite faces) as repeated measures and the cognitive reserve group as between-participants measure. In addition (see Herzig et al., 2010, for detailed rationale) we performed a mixed-samples ANOVA on mean reaction time for sex decisions with face type (correct decisions for WF, LF decisions, RF decisions) as a repeated measure and the cognitive reserve group as a between-participants measure. Post-hoc tests correcting for multiple comparisons were performed using Tukey HSD tests. Effect sizes (*partial* η^2) are reported for all ANOVA results.

We also tested whether, within each cognitive reserve group, the tasks resulted in lateralised performance at all (Herzig et al., 2010; Mason &

Claridge, 1999; Mohr et al., 2006) using conventional laterality indices (Marshall, Caplan, & Holmes, 1975) by subtracting inferior performance from superior performance, and dividing this difference by its sum. Accordingly, positive values indicated an advantage of the normally dominant hemisphere (LDT: left hemisphere/LH; FDT: right hemisphere/RH), and negative values an advantage of the normally sub-dominant hemisphere. In order to obtain indices that would be comparable in this respect, the index for (i) accuracy in the LDT, and (ii) reaction times in the FDT was $[(RVF-LVF)/(LVF+RVF)]*100$, while the index for reaction times in the LDT was $[(LVF-RVF)/(LVF+RVF)]*100$. For accuracy in the FDT we needed only the percentage of LF decisions (as LF and RF decisions added up to 100%). These indices were calculated for each cognitive reserve group separately, with one sample *t*-tests against chance level (zero for both LDT indices and the FDT RT index, 50% for the FDT LF percentage index).

To establish an effect of schizotypy over and above age, cognitive reserve (average mark), substance-CB, and stress (perceived stress scale, stress response inventory) on hemispheric asymmetry, we performed hierarchical regressions as follows. Age was entered in the first step (Amirkhan & Auyeung, 2007; Brallier, Palm, & Gilbert, 2007; see also Table 3), cognitive reserve (average mark) was entered in the second step, substance-CB in the third step, stress measures (perceived stress scale, stress response inventory) in the fourth step, and schizotypy (UnEx scores, CogDis scores, IntAn scores, Imp scores) in the fifth step. Thus five blocks of predictors were entered in nested blocks, meaning that each subsequent block contained all prior predictors and additional predictors from the current block. In line with Fridberg, Vollmer, O'Donnell, and Skosnik (2011), the presentation of results will be restricted to significant ΔR^2 values and their corresponding significant coefficients. We considered multi-collinearity between independent variables negligible, because all tolerance values were above .2 (Menard, 1995), and all independent variables were mean-centred. Separate regression analyses were applied to the same dependent variables included in the ANOVAs described above (see also Herzig et al., 2010). Kolmogorov-Smirnov tests for the groups separately revealed normal distribution for all behavioural measures and questionnaire scores. All *p*-values were two-tailed and the α -level was set at .05.

RESULTS

Participants and self-report questionnaires

Most participants did not report any particularly stressful previous life events ($n = 44$). Self-report measures were compared to previous normative samples via calculations of Cohen's *d* (1992) with values of $\pm 0.2/\pm 0.5/\pm 0.8$ being indicative of a small/medium/large effect size, respectively (see

TABLE 1
Means, *SD*, and effect sizes (Cohen’s *d*), comparing the normative sample values with our sample

Questionnaire	Published norms by	Norm values			Our sample (<i>N</i> = 58)		
		<i>m</i>	<i>SD</i>	<i>N</i>	<i>m</i>	<i>SD</i>	Cohen’s <i>d</i>
O-LIFE: UnEx ^a	Mason et al. (2005)	3.35	2.92	2072	3.67	1.89	−0.11
O-LIFE: CogDis ^b	Mason et al. (2005)	4.42	2.90	2094	6.16	2.55	−0.60
O-LIFE: IntAn ^c	Mason et al. (2005)	2.46	2.01	2073	1.40	1.41	0.53
O-LIFE: Imp ^d	Mason et al. (2005)	2.60	1.99	2098	2.78	1.52	−0.09
PSS ^e	Ramirez & Hernandez (2007)	21.90	7.03	365	28.21	8.00	−0.88
SRI ^f	Koh et al. (2001)	68.50	23.40	215	41.96	20.79	1.16

^aUnusual experiences; ^bCognitive Disorganisation; ^cIntrovertive Anhedonia; ^dImpulse non-conformity; ^ePerceived Stress Scale; ^fStress Response Inventory.

TABLE 2
Means and *SD* for age and self-report questionnaire scores for the whole sample,
and for the two cognitive reserve groups separately

	Total (N = 58)		High (N = 31)		Low (N = 27)			
Dependent variable	<i>Mean</i>	<i>SD</i>	<i>Mean</i>	<i>SD</i>	<i>Mean</i>	<i>SD</i>	<i>t</i> (1, 56)	<i>p</i>
Age	20,33	2,35	20,97	2,87	19,59	1,25	2,42	0,02
UnEx ^a	3,67	1,89	3,26	1,71	4,15	1,99	-1,83	0,07
CogDis ^b	6,16	2,55	5,55	2,59	6,85	2,35	-1,99	0,05
IntAn ^c	1,40	1,41	1,42	1,29	1,37	1,57	0,13	0,90
Imp ^d	2,78	1,52	2,71	1,47	2,85	1,61	-0,35	0,73
SCB ^e	12,38	8,60	11,61	7,41	13,26	9,87	-0,72	0,47
PSS ^f	28,21	8,00	27,52	7,31	29,00	8,80	-0,70	0,49
SRI ^g	41,96	20,79	39,19	18,34	45,13	23,23	-1,09	0,28

The results of the *t*-tests are also presented. ^aUnusual experiences; ^bCognitive Disorganisation; ^cIntroversive Anhedonia; ^dImpulse non-conformity; ^eSubstance-based compensatory behaviours; ^fPerceived Stress Scale; ^gStress Response Inventory.

Table 1). These comparisons showed that the present sample was largely comparable with normative samples, apart from CogDis scores and perceived stress scale scores that were elevated and IntAn scores and stress response inventory scores that were reduced in our as compared to the normative sample (see Table 1).

Regarding the cognitive reserve groups, the median-split (median of 65%) on the average mark (in percent) resulted in 31 students (23 women) belonging to the high cognitive reserve group (marks $\geq 65\%$, range 65–80%, $m = 68.55$, $SD = 4.12$), and 27 students (19 women) belonging to the low cognitive reserve group (marks $< 65\%$, range 40–64%, $m = 56.44$, $SD = 6.70$). Separate *t*-tests on age, schizotypy scores, substance-CB scores, perceived stress scale scores, and stress response inventory scores with cognitive reserve group as a between-participants measure showed that the high as compared to low cognitive reserve group was older, and had lower CogDis scores (Table 2). None of the other group comparisons was significant (Table 2).

Correlations between self-report questionnaire scores

The correlations between schizotypy measures showed positive correlations between UnEx, Imp, and CogDis scores, but no correlations with IntAn scores (Table 3). The stress response inventory scores were positively correlated with perceived stress scale scores (Table 3). Correlations between schizotypy and stress-relevant measures showed that increasing UnEx scores, CogDis scores, and Imp scores all related to increasing perceived stress scale scores and stress response inventory scores. Substance-CB scores were

TABLE 3
Pearson correlations between scale scores

	UnEx	CogDis	IntAn	Imp	SCB	PSS	SRI	Average Mark
UnEx ^a	1.00	.37**	−0.02	.36**	0.18	.32*	.42***	−0.18
CogDis ^b	.37**	1.00	0.02	.44***	.46***	.45***	.56***	− 0.22†
IntAn ^c	−0.02	0.02	1.00	0.24†	−0.01	0.05	0.01	0.23†
Imp ^d	.36**	.44***	0.24†	1.00	0.13	.33*	.36**	−0.08
SCB ^e	0.18	.46***	−0.01	0.13	1.00	0.19	0.24†	−0.14
PSS ^f	.32*	.45***	0.05	.33*	0.19	1.00	.69***	−0.20
SRI ^g	.42***	.56***	0.01	.36**	0.24†	.69***	1.00	− 0.23†
Average mark	−0.18	− 0.22†	0.23†	−0.08	−0.14	−0.20	− 0.23†	1.00

^aUnusual experiences; ^bCognitive Disorganisation; ^cIntroverted Anhedonia; ^dImpulse non-conformity; ^eSubstance-based compensatory behaviours; ^fPerceived Stress Scale; ^gStress Response Inventory. [†] $p \leq .10$; * *significant at* $p \leq .05$; ** *significant at* $p \leq .01$; *** *significant at* $p \leq .001$.

positively correlated with CogDis scores (Table 3). An inverse relationship between the extent of schizotypal traits and IQ is sometimes reported (e.g., Matheson & Langdon, 2008; Noguchi et al., 2008), even though IQ is generally believed to be spared in schizotypy (Raine, 2006). In this sample none of the correlations between schizotypy and average mark was significant (Table 3).

Hemispheric asymmetry tasks

LDT. The ANOVA on accuracy showed that performance was superior for RVF as compared to LVF presentations, $F(1, 56) = 16.97$, $p < .001$, *partial* $\eta^2 = .23$ (see Table 4). The interaction between group and visual field, $F(1, 56) = .27$, $p = .60$, *partial* $\eta^2 < .01$, and the main effect of group, $F(1, 56) = 1.21$, $p = .28$, *partial* $\eta^2 = .02$ (see Table 4) yielded no significant result. The ANOVA on reaction times showed no significant result (Table 4), neither for the main effect visual field, $F(1, 56) = .08$, $p = .78$, *partial* $\eta^2 < .01$, nor the main effect group, $F(1, 56) = .14$, $p = .71$, *partial* $\eta^2 < .01$, nor the interaction visual field * group, $F(1, 56) = 1.84$, $p = .18$, *partial* $\eta^2 = .03$.

The comparisons against chance level for the laterality indices were significant for percent accuracy, $t(57) = 3.66$, $p < .001$, but not for reaction

TABLE 4
Mean and SD of lateralised task performance for the total sample and the two groups separately

		<i>All (N = 58)</i>		<i>High (N = 31)</i>		<i>Low (N = 27)</i>	
		<i>Mean</i>	<i>SD</i>	<i>Mean</i>	<i>SD</i>	<i>Mean</i>	<i>SD</i>
LDT ^a	LDT LVF ^c % ^d	62.14	20.99	64.85	21.28	59.03	20.60
	LDT RVF ^c %	74.43	15.87	75.67	14.60	72.99	17.38
	LDT NoW ^f %	72.05	10.56	73.19	10.63	70.76	10.52
	LDT index ^g %	10.34	21.52	9.50	20.18	11.30	23.32
	LDT LVF RT	695.71	108.57	691.93	92.85	700.05	125.93
	LDT RVF RT	693.24	102.81	705.59	104.76	679.06	100.58
	LDT NoW RT	834.16	133.27	843.38	118.29	823.57	150.25
FDT ^b	LDT index RT	0.10	6.74	-0.88	6.96	1.23	6.41
	FDT LF ^h %	59.03	9.89	61.37	9.12	56.34	10.22
	FDT WF ⁱ %	90.82	7.03	91.29	6.98	90.28	7.18
	FDT LF RT	678.82	124.37	683.47	132.41	673.49	116.74
	FDT RF ^j RT	702.28	145.31	713.25	155.59	689.69	134.35
	FDT WF RT	597.47	98.74	601.31	104.21	593.05	93.84
	FDT index RT	1.47	4.01	1.86	4.34	1.02	3.64

^aLexical decision task; ^bFacial decision task; ^cLeft visual field; ^dPercentage correct; ^eRight visual field; ^fTwo non words displayed on either side of the screen; ^gLaterality index; ^hLeft face decisions; ⁱWhole face decisions; ^jRight face decisions.

times, $t(57) = .12$, $p = .91$. The difference remained stable when calculated for the cognitive reserve groups separately [high cognitive reserve group: accuracy: $t(30) = 2.62$, $p = .01$; reaction time: $t(30) = -.70$, $p = .49$; low cognitive reserve group: accuracy: $t(26) = 2.52$, $p = .02$; reaction time: $t(26) = .99$, $p = .33$ (Table 4)]. The positive laterality indices point to a RVF (LH) advantage in the LDT.

FDT. The ANOVA on percentage of sex decisions showed a main effect for face type with sex decisions for WF being more accurate than the proportion of LF decisions, $F(1, 56) = 443.85$, $p < .001$, *partial* $\eta^2 = .89$ (Table 4). The interaction between face type and group, $F(1, 56) = 1.76$, $p = .19$, *partial* $\eta^2 = .03$, and the main effect for group, $F(1, 56) = 3.43$, $p = .07$, *partial* $\eta^2 = .06$, were not significant. The ANOVA on reaction times showed a significant main effect for face type, $F(2, 112) = 76.63$, $p < .001$, *partial* $\eta^2 = .58$, due to slower responses for both RF decisions ($p < .001$) and LF decisions ($p < .001$) as compared to WF decisions (Table 4). However, participants were equally fast for LF and RF decisions ($p = .15$). The main effect for cognitive reserve, $F(1, 56) = .20$, $p = .66$, *partial* $\eta^2 < .01$, and the interaction between cognitive reserve and visual field were both not significant, $F(2, 112) = .45$, $p = .61$, *partial* $\eta^2 = .01$. The comparisons against chance level for the laterality indices showed a significant finding for percent LF decisions, $t(57) = 6.95$, $p < .001$ (see Table 4) as well as the reaction time index, $t(57) = 2.79$, $p = .01$. The analogue comparisons for the cognitive reserve groups separately were significant for the high, $t(30) = 6.94$, $p < .001$, and low, $t(26) = 3.23$, $p < .01$, group on percent LF decisions, but only for the high group, $t(30) = 2.39$, $p = .02$ —low group: $t(26) = 1.46$, $p = .16$ (Table 4)—on the reaction time index. Percent LF decisions above 50% represent a LF (RH) bias.

Regressions: Severity of compensatory behaviours, stress, and schizotypy as predictors of performance in the LDT and FDT

LDT. Neither age nor cognitive reserve significantly predicted variance in the LDT (see Table 5). Compensatory behaviours explained a significant amount of variance in the LDT outcome variables, with increasing substance-CB scores predicting an increase in LVF% and a decrease in the index% (Table 5; see also Herzig et al., 2010). Adding the two stress scores in the fourth step only helped to explain variance in the reaction times for correct lexical decisions in the RVF. More specifically, faster RVF reaction times were predicted by increasing perceived stress scale scores (Table 5). Schizotypy explained a significant amount of variance in the reaction time index only with higher CogDis scores enhancing the typical LH-dominant response pattern (see Table 5).

TABLE 5
Beta-weights and ΔR^2 for the LDT outcome variables for the whole sample

Outcome variables	Step	Significant predictor	β -value	Total R^2	ΔR^2	F for ΔR^2
LVF ^a % ^b	3	SCB ^d	0.39**	0.18**	0.14**	9.20**
Index ^c	3	SCB	-0.29*	0.10*	0.08*	4.80*
RVF RT	4	PSS ^c	-0.38*	0.14*	0.13*	3.97*
RT Index	5	CogDis ^f	0.39*	0.28*	0.21*	3.45*

Beta-weights and ΔR^2 for the LDT outcome variables for the whole sample, assessing the effect of schizotypy (step 5) on LDT performance on top of age (step 1), cognitive reserve (step 2), compensatory behaviours (step 3) and stress (step 4). Only significant ΔR^2 -values and their corresponding significant coefficients (β -values) are reported.

*significant at $p \leq .05$; **significant at $p \leq .01$; ***significant at $p \leq .001$

^aLeft visual field; ^bPercentage correct; ^cLaterality index; ^dSubstance-based compensatory behaviours; ^ePerceived Stress Scale; ^fCognitive Disorganisation.

TABLE 6
Beta-weights and ΔR^2 for the FDT outcome variables for the whole sample

Outcome variable	Step	Significant predictor	β -value	Total R^2	ΔR^2	F for ΔR^2
RT index ^a	1	Age	-0.26*	0.07*	0.07*	4.20*
	4	PSS ^b	0.39*	0.21*	0.13*	4.19*
	4	SRI ^c	-0.52**	SAA^d	SAA	SAA

Beta-weights and ΔR^2 for the FDT outcome variables for the whole sample, assessing the effect of schizotypy (step 5) on FDT performance on top of age (step 1), cognitive reserve (step 2), compensatory behaviours (step 3) and stress (step 4). Only significant ΔR^2 -values and their corresponding significant coefficients (β -values) are reported.

*significant at $p \leq .05$; **significant at $p \leq .01$; ***significant at $p \leq .001$.

^aLaterality index; ^bPerceived Stress Scale; ^cStress Response Inventory; ^dSame value as above.

FDT. As Table 6 indicates, only age and stress predicted an additional significant amount of variance on top of the other variables in the FDT RT index. More specifically, increasing age predicted a reduction in the typical RH bias for face processing. The stress measures predicted variance in the FDT RT index in opposite directions: Whereas perceived stress predicted a stabilisation of the RH face processing bias, the stress response predicted a decrease in RH bias (see Table 6).

DISCUSSION

Relative cognitive impairments along the schizophrenia spectrum might be mediated by stress rather than those processes related to the symptoms of the

schizophrenia spectrum. To test this idea, we measured functional hemispheric asymmetry in a student population varying in schizotypal thoughts around their end of year exam period. We assessed hemispheric asymmetry, because its attenuation has been related to enhanced psychotic (clinical populations) or psychotic-like (schizotypal) thinking styles (Broks, 1984; Brugger et al., 1993; Mason & Claridge, 1999; Reichenberg & Harvey, 2007; Suzuki & Usher, 2009). To account for individual differences in perceived and experienced stress (Dallman, 2010; Ensel & Lin, 2004; Finkelstein et al., 2006; Goeders, 2003; Krause, Goldenhar, Liang, Jay, & Maeda, 1993; Scarmeas & Stern, 2003), we assessed perceived stress and stress response (S. Cohen et al., 1983; Koh et al., 2001), stress-related compensatory behaviours (substance use), and potential cognitive reserve. If stress is a more important predictor of hemispheric asymmetry than schizotypy, we expected that higher stress levels and higher stress-related compensatory behaviours, in particular in those with less cognitive reserve (relative lower marks), relate to reduced hemispheric asymmetry for function.

Our data supported these predictions only partially, i.e., enhanced substance-CB contributed to a reduced LH dominance for language (see also Herzig et al., 2010) but enhanced perceived stress was associated with increased hemispheric asymmetry. We did not find any consistent mediation of performance by cognitive reserve or the stress response. Regarding schizotypy, increasing CogDis scores were associated with enhanced LH language dominance. Before discussing the relevance of these findings, we would like to point out that we tested a seemingly representative population because we first observed the commonly reported LH bias in the LDT and the RH bias in the FDT (Brugger et al., 1993; Mason & Claridge, 1999; Regard, Landis, & Graves, 1985) and second, that self-report measures were largely comparable to normative values, apart from some scores that were relatively higher in our sample for the perceived stress scale and CogDis subscale and relatively lower for the stress response inventory and IntAn subscale.

Hemispheric asymmetry and its relation to substance-CB

A major finding was that increased substance-CB related to a reduced LH language dominance supporting previous notions that drug use relates to a cognitive profile resembling the one seen along the schizophrenia spectrum, namely an attenuated hemispheric asymmetry pattern (Bleich-Cohen et al., 2009; Broks, 1984; Brugger et al., 1993; Kucharska-Pietura et al., 2002; Løberg et al., 2006; Mason & Claridge, 1999; Mitchell & Crow, 2005; Mohr et al., 2005; Phillips & David, 1997; Sommer et al., 2001; Suzuki & Usher, 2009). Thus an attenuated hemispheric asymmetry seems related not only to enhanced nicotine dependence in healthy

populations (Herzig et al., 2010; Hahn, Pogun, & Güntürkün, 2010) and patients with schizophrenia (Hahn et al., 2011; Hahn et al., 2010), but might relate to a generally enhanced drug use. Indeed, participants in these latter nicotine studies are likely to have consumed other drugs (e.g. licit and/or illicit) as well (Degenhardt, Hall, & Lynskey, 2001; Martinez-Ortega, Jurado, Martinez-Gonzalez, & Gurpegui, 2006), stressing the possibility that the present RH shift in language processing with increasing substance-CB is not restricted to a particular substance (Herzig et al., 2010). Enhanced substance-CB as a potential indicator of a “pathological” profile would also be supported by the observation that elevated substance-CB predicted more cognitive disorganisation. This schizotypy subscale was also linked with lower cognitive reserve and higher perceived stress in the present and independent samples (e.g. Ventura et al., 2010; Cuesta & Peralta, 1995; Walker et al., 2008) and with deteriorating mental health in the long term (Goulding & Ödéhn, 2009; Schofield & Claridge, 2007).

As a final note, we found the relationship between attenuated hemispheric asymmetry and enhanced substance dependence only for the LDT (but see Herzig et al., 2010), potentially because language is the cognitive function that most reliably yields lateralised findings in healthy populations (Hugdahl & Westerhausen, 2010; Resnick, Lazar, Gur, & Gur, 1994).

Hemispheric asymmetry and its relation to stress

Perceived stress was associated with a LH shift in the LDT and FDT, whereas the stress response predicted a reduction in RH face processing for one outcome variable only. Such a LH shift, in particular for the more reliable LH language functions, may point to a favourable cognitive profile with enhanced perceived stress. Some authors argue that stress is advantageous for general survival (for an overview see Lupien, Maheu, Tu, Fiocco, & Schramek, 2007) or performance abilities (Lewis, Nikolova, Chang, & Weekes, 2008; McEwen & Sapolsky, 1995; Yerkes & Dodson, 1908). Supporting different, opposing conclusions for results on substance-CB above, and stress-measures here, these two measures were not related with each other. By inference, perceived stress may be related to a relatively “healthy” pattern of brain functioning, while the need to engage in substance-CB during times of enhanced stress (exams) may be linked to a relatively “unhealthy” pattern of brain functioning (Herzig et al., 2010).

Hemispheric asymmetry and its relation to schizotypy

Increasing CogDis scores were associated with an increase in LH language dominance for the RT measure. This result is contrary to our initial

prediction and might thus be spurious. Caution is also warranted when considering that CogDis relates to poor verbal abilities (Chan et al., 2011; Cuesta & Peralta, 1995; Lucas et al., 2004; Stefanis et al., 2006). Taking into account that high CogDis is indicative of a rather disadvantageous cognitive profile (see above), and that it was related to elevated stress perception and drug use, we are hesitant to conclude that CogDis favours the most common and, by inference, “normal” cognitive profile. We would like to speculate instead that CogDis may be particularly sensitive to environmental changes, based on the following reasoning. Previous findings investigating the link between CogDis and language laterality have been inconsistent in the literature. A reduction in language laterality is most consistently reported from samples taken from the general population (Mason & Claridge, 1999; Suzuki & Usher, 2009). Inconsistent findings on the relationship between CogDis and language laterality, on the other hand, emerge frequently for student samples, be it decreases in LH processing (Herzig et al., 2010; Kravetz, Faust, & Edelman, 1998), increases in LH processing (see Liouta, Smith, & Mohr, 2008; and this study), or no differences between hemispheres (Nunn & Peters, 2001). Purportedly, as compared to the individuals from the general population, students have unique features not observed in the general population that predispose them to fluctuations in laterality patterns, such as during confined periods of exam stress. We are unaware of any published report linking CogDis, stress, and laterality, but do know that stress is associated with psychotic-like symptoms (e.g., Lee & Schepp, 2009; van Os, et al., 2009; Walker & Diforio, 1997; Walker et al., 2008). It may be the case that relatively elevated CogDis might turn exam periods into particularly stressful situations. We actually found that higher CogDis was associated with higher perceived stress. This enhanced stress may elevate revision time in students high on CogDis exposing them to more verbal stimuli and consequently increase LH language laterality (Marcel, Katz, & Smith, 1974; Rutherford, 2006; Wey, Cook, Landis, Regard, & Graves, 1993). On the other hand, during periods of low exam stress, low revision time, and lower exposure to verbal material CogDis might have related to the expected hemispheric asymmetry pattern, i.e., a reduced LH language dominance. The present argumentation is admittedly highly conjectural but constructive enough to trigger future studies in which participants’ situational variables are accounted for.

The influence of drug use on the link between schizotypy and cognition

Substance use is elevated along the schizophrenia spectrum for a large variety of drugs, such as (i) tobacco use in schizotypy (Esterberg et al., 2007;

Williams et al., 1996) and schizophrenia patients (de Leon, Diaz, Rogers, Browne, & Dinsmore, 2002), (ii) cannabis in schizotypy (Barkus, Stirling, Hopkins, & Lewis, 2006; Skosnik, Spatz-Glenn, & Park, 2001; Williams, Wellman, & Rawlins, 1996) and schizophrenia (Archie et al., 2007; Moore et al., 2007), (iii) amphetamines/stimulants in schizotypy (see Curran & Morgan, 2000 for overview) and schizophrenia (Barkus & Murray, 2010), and (iv) caffeine in schizotypy (Jones & Fernyhough, 2009) and schizophrenia (Gurpegui, Aguilar, Martinez-Ortega, Diaz, & de Leon, 2004). In our present study it was not schizotypy but substance use that was associated with a reduced LH language dominance, indicating that substance use might be the major determinant to measures of cognitive functioning. The present study and its design are unsuited to inferring which variable (schizotypy, substance use, or yet unspecified variables) is the most likely to cause a reduced LH language dominance. Reports on legal or illegal substance are still uncommon in the schizotypy literature when investigating the effects of psychotic(like) thinking on hemispheric asymmetry (Broks, 1984; Brugger et al., 1993; Mason & Claridge, 1999; Mohr et al., 2005; Suzuki & Usher, 2009), despite the recent evidence that the extent of substance use is important in this regard (Hahn et al., 2011, 2010; Herzig et al., 2010). The necessity to control for substance use is not exclusive to the study of hemispheric asymmetry but is relevant to cognitive functions more broadly (e.g. Rodriguez-Jimenez et al., 2010). Only by taking the influence of substance use on cognitive functions in schizotypy studies more seriously will we find out whether our focus on schizotypal symptoms has been overvalued.

Limitations

While we have aimed to control and account for important variables to test our study question, we are aware of shortcomings and possible improvements, such as (i) the pre-selection of right-handed participants, (ii) inferring cognitive reserve from the current mark, (iii) assessing substance-CB without knowing anything about the substance(s), (iv) unbalanced sex distribution. Concerning the first point we pre-selected participants for their right-handedness as is common in experiments on hemispheric asymmetry in general (Bourne, 2006) and on schizotypy in particular (e.g., Mason & Claridge, 1999; Mohr et al., 2005; Suzuki & Usher, 2009) and general hemispheric asymmetry (Bourne, 2006). However, if we acknowledge reduced right-handedness in schizophrenia (Dragovic & Hammond, 2005) and schizotypy (Chapman, Grimshaw, & Nicholls, 2011; Somers, Sommer, Boks, & Kahn, 2009), we are likely to exclude study-relevant participants. Future studies should investigate a more representative sample that represents a wider range of hand preferences (e.g., Shaw, Claridge, & Clark, 2001; Somers et al., 2009). Concerning the second point we tested a

high-performing student group, because all had to have three As (in A-level exams) when admitted to the local Psychology course. Using participants from a wider educational background might yield more systematic relationships between task performance and study performance (i.e., marks) than the one observed here. Concerning the third point the substance-CB questionnaire did not ask about specific substances that could nevertheless have yielded different effects on hemispheric asymmetry. For instance, some of the substances might have stress-alleviating properties (e.g., anxiolytic medication), having counteracted otherwise detrimental effects of stress on cognition. The advantage of the present questionnaire is its lowered risk of socially desirable responding. Moreover, as argued above, results from the questionnaire replicated previous findings on specific drugs such as nicotine. Concerning the fourth point, studies on laterality and schizotypy have frequently observed effects in male populations only (see Hausmann & Güntürkün, 2000 for overview; McGlone, 1980), hence we cannot exclude stronger and more consistent findings if we had excluded female participants. However, due to unequal sex distribution in the current sample we could not further account for this factor.

Conclusions

We here aimed to disentangle whether stress and stress-related behaviours or schizotypal symptoms mediate reduced hemispheric asymmetry for function as formerly reported along the schizophrenia spectrum. Results from two lateralised tasks (LDT, FDT) were not fully supportive of either perspective, but revealed that stress, stress-related behaviour, and schizotypal symptoms contributed differently to cognitive performance. Most importantly, our findings indicated that (i) a RH shift in language dominance might be explained by enhanced substance dependence, and less so by individuals' schizotypal features, (ii) perceived stress related to a LH shift, and (iii) inconsistent findings on laterality and schizotypy might be population and context dependent.

In practical terms, the implications of these results are that future studies assessing the link between schizotypy and cognition should control for legal and illegal substance use, in which participants' life situations are measured (stressful life situation or not), and whether students or non-students of various educational backgrounds are targeted. We also argue that the current implications are relevant not only to studies on hemispheric asymmetry, but to cognition more generally and frontal lobe functioning more particularly (Buchsbaum et al., 2002; Laurent et al., 2001; Reichenberg & Harvey, 2007; Tsakanikos & Claridge, 2005).

NOTE

1. We also assessed verbal and figural fluency performance. For economy of presentation these findings will be presented elsewhere. Data and findings from the fluency tasks can also be requested from the corresponding author.

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APPENDIX
Compensatory behaviour questionnaire

Please think about a rewarding habit you cannot imagine living without. This habit should be substance-based (like caffeine, chocolate, nicotine or illicit drugs).

Please answer the following questions regarding your habit in the past month, using the scales below.

Substance-based habit

Do you need the habit in order to perform/ feel well?	Never	Sometimes	Often	Nearly always	Always
Do you find yourself thinking about when you will next be able to perform your habit?	Never	Sometimes	Often	Nearly always	Always
Does your pattern of performing your habit interfere with your social life?	Never	Sometimes	Often	Nearly always	Always
Do you plan your days around performing your habit?	Never	Sometimes	Often	Nearly always	Always
Do you feel that your need for performing this habit is too strong to control?	Never	Sometimes	Often	Nearly always	Always
Do you feel agitated or irritable when you cannot perform your habit?	Never	Sometimes	Often	Nearly always	Always
Do you need your habit in order to relieve stress?	Never	Sometimes	Often	Nearly always	Always
Is performing your habit more important than anything else you might do during the day?	Never	Sometimes	Often	Nearly always	Always
Has performing your habit become a problem for you?	Never	Sometimes	Often	Nearly always	Always
Do you feel you cannot cope with life without that habit?	Never	Sometimes	Often	Nearly always	Always
Are you feeling more positive about yourself and/ or life after performing your habit?	Never	Sometimes	Often	Nearly always	Always
Do you perform your habit in higher amounts/ frequency than intended?	Never	Sometimes	Often	Nearly always	Always
Does your habit interfere with your study?	Never	Sometimes	Often	Nearly always	Always
Do you find it difficult to cope with life without your habit?	Never	Sometimes	Often	Nearly always	Always
Do you feel guilty about performing your habit?	Never	Sometimes	Often	Nearly always	Always
Do you have little energy for your family and friends due to your habit?	Never	Sometimes	Often	Nearly always	Always